

理想气体的热力学函数

$$1. \begin{cases} C_v = \left(\frac{\partial U}{\partial T} \right)_v = \frac{dU(T)}{dT} = C_v(T) \\ C_p = \left(\frac{\partial H}{\partial T} \right)_p = \frac{dH(T)}{dT} = C_p(T) \end{cases}$$

(1) $U(T) = \int C_v(T) dT + U_0$

由 $du = Tds - pdv \rightarrow$

$$ds = \frac{1}{T} du + \frac{p}{T} dv$$

$$= \frac{dT}{T} C_v + NR \frac{dv}{v}$$

$$\boxed{pV = NRT}$$

$$ds = d \left[\int \frac{C_v(T)}{T} dT \right] + d[NR \ln V]$$

$$d \left[S - \int \frac{C_v(T)}{T} dT - NR \ln V \right] = 0$$

$$S - \int \frac{C_v(T)}{T} dT - NR \ln V = S_0$$

$$S = \int \frac{C_v(T)}{T} dT + NR \ln V + S_0 = S(T, V)$$

利用 $C_p - C_v = NR$ $pV = NRT$

$$S = \int \frac{C_p(T)}{T} dT - \int \frac{NR}{T} dT + NR \ln V + S_0$$

$$= \int \frac{C_p(T)}{T} dT - NR \ln T + NR \ln V + S_0$$

$$= \int \frac{C_p(T)}{T} dT - NR \ln \frac{P}{NR} + S_0$$

$$= \int \frac{C_p(T)}{T} dT - NR \ln P + S'_0 = S(T, P)$$

*例 $H = \int C_p(T) dT + H_0$

$$= U + PV = \int C_v(T) dT + PV + U_0$$

$$= \int C_p(T) dT - \int NR dT + PV + U_0$$

$$= \int C_p(T) dT - NRT + PV + NRT_0 + U_0$$

$$= \int C_p(T) dT + U_0 + P_0 V_0$$

$$= \int C_p(T) dT + H_0$$

自由能 $F = U - TS$

$$= \int C_v dT - T \int \frac{C_v}{T} dT - NRT \ln V + U_0 - TS_0$$

利用 $xy = \int x dy + \int y dx$

令 $x = \frac{1}{T}$ $y = \int C_v dT$

前两项为 $T \left[\frac{1}{T} \int C_v dT - \int \frac{C_v}{T} dT \right]$

$$= T \left[xy - \int x dy \right]$$

$$= T \int \int C_v dT d\left(\frac{1}{T}\right)$$

$$= -T \int \frac{\int C_v dT}{T^2} dT = -T \int \frac{dT}{T^2} \int C_v dT$$

$$F = -T \int \frac{dT}{T^2} \int C_v dT - NRT \ln V + U_0 - TS_0$$

$$G = U - TS + PV = H - TS$$

$$= \int C_p dT + H_0 - T \left[\int \frac{C_p}{T} dT - NR \ln p + S_0 \right]$$

$$= \int C_p dT - T \int \frac{C_p}{T} dT + NRT \ln p + H_0 - TS_0$$

$$\text{令 } x = \frac{1}{T} \quad y = \int C_p dT$$

$$\text{前两项} = T \left[\frac{1}{T} \int C_p dT - \int \frac{C_p}{T} dT \right]$$

$$= T [xy - \int x dy] = T \int y dx$$

$$= -T \int \frac{dT}{T^2} \int C_p dT$$

$$G = -T \int \frac{dT}{T^2} \int C_p dT + NRT \ln p + H_0 - T$$

2. 化学势：化学纯物质 1mol 的吉布斯函数
用 μ 表示

$$\mu = U - TS + PV = h - TS$$

$$= -T \int \frac{dT}{T^2} \int C_p dT + RT \ln p + h_0 - TS_0'$$

$$u = \frac{G}{N} \quad u = \frac{U}{N} \quad s = \frac{S}{N} \quad v = \frac{V}{N}$$

$$h = \frac{H}{N}$$

特性函数

1. 可以证明, 在独立变量适当选择之下, 只要知道一个热力学函数就可以确定均匀系的全部平衡性质. 这个函数称为特性函数.

独立变量	特性函数	微分方程
(S, V)	$U(S, V)$	$dU = TdS - PdV$
(S, P)	$H(S, P)$	$dH = TdS + VdP$
(T, V)	$F(T, V)$	$dF = -SdT - PdV$
(T, P)	$G(T, P)$	$dG = -SdT + VdP$

$P-T-V$ 系统

3. 证明 $U(S, V)$ 是特性函数.

由 $U(S, V)$ 确定三个基本热力学函数.
(以可测量量为变量)

$$dU = TdS - PdV$$

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

$$P = - \left(\frac{\partial U}{\partial V} \right)_S$$

$$U = U(S, V) \Rightarrow T = T(S, V) \Rightarrow$$

$$S = S(T, V)$$

$$\text{由 } p = -\left(\frac{\partial U}{\partial V}\right)_S = p(S, V)$$

$$= p(S(T, V), V) = p(T, V)$$

$$\text{将 } S = S(T, V) \text{ 代入 } U = U(S, V)$$

$$\text{得 } U = U(T, V)$$

由 $U = U(S, V)$ 确定了以 (T, V) 为变量的

U, p, S

$$\text{例: 已知 } U = C S^{\frac{4}{3}} V^{-\frac{1}{3}} \quad (C \text{ 为常数})$$

$$T = \left(\frac{\partial U}{\partial S}\right)_V = \frac{4C}{3} \left(\frac{S}{V}\right)^{\frac{1}{3}}$$

$$S = \left(\frac{3T}{4C}\right)^3 V \Rightarrow \frac{S}{V} = \left(\frac{3T}{4C}\right)^3$$

$$p = -\left(\frac{\partial U}{\partial V}\right)_S = \frac{C}{3} S^{\frac{4}{3}} V^{-\frac{4}{3}} = \frac{C}{3} \left(\frac{S}{V}\right)^{\frac{4}{3}}$$

$$= \frac{C}{3} \left(\frac{3T}{4C}\right)^4 \propto \frac{a}{3} T^4$$

$$U = C \left[\left(\frac{3T}{4C}\right)^3 V \right]^{\frac{4}{3}} V^{-\frac{1}{3}} = C \left(\frac{3T}{4C}\right)^4 V$$

4. 证明 $F(T, V)$ 是特性函数.

$$dF = -SdT - PdV$$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V \Rightarrow S = S(T, V)$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_T \Rightarrow P = P(T, V)$$

$$F = U - TS$$

$$\begin{aligned} U &= F + TS = F - T\left(\frac{\partial F}{\partial T}\right)_V = -T^2 \frac{\partial}{\partial T} \left(\frac{F}{T}\right) \\ &= F(T, V) + TS(T, V) = U(T, V) \end{aligned}$$

5. 证明 $G(T, P)$ 是特性函数.

$$dG = -SdT + VdP$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_P \Rightarrow S = S(T, P)$$

$$V = \left(\frac{\partial G}{\partial P}\right)_T \Rightarrow V = V(T, P)$$

$$G = U - TS + PV$$

$$U = G + TS - PV$$

$$= G(T, P) + TS(T, P) - PV(T, P)$$

$$= U(T, P)$$

$$H = G - T\left(\frac{\partial G}{\partial T}\right)_P = -T^2 \frac{\partial}{\partial T} \left(\frac{G}{T}\right)$$

吉布斯-亥姆霍兹方程

6. 证明 $H = H(S, P)$ 是等压性函数。

$$dH = TdS + VdP$$

$$T = \left(\frac{\partial H}{\partial S} \right)_P \Rightarrow T = T(S, P) \Rightarrow S = S(T, P)$$

$$V = \left(\frac{\partial H}{\partial P} \right)_S \Rightarrow V = V(S, P) = V(S(T, P), P) \\ = V(T, P)$$

$$U = H - PV = H(S(T, P), P) - PV(T, P)$$

$$= H(T, P) - PV(T, P)$$

$$= U(T, P)$$